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Leveraging Hidden Markov Model with Window Filtering Technique for Unsupervised Facies Detection in Well Logs

Lundi Kusuma and Alwin Djamaoeddin, Saudi Aramco, Dhahran, Saudi Arabia

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Abstract

Facies identification from basic well logs is essential for optimizing well operation or constructing reservoir model. The sequential nature of well log data makes the Hidden Markov Model (HMM) a suitable unsupervised machine learning approach for clustering. This study investigates the effectiveness of HMM combined with window filtering for unsupervised facies detection and compares its performance with the widely used K-Means Clustering (KMC) method.

Synthetic gamma ray (GR), density (DEN), and neutron (NEU) logs, along with geologist-guided facies in a channel depositional environment, were used as input data. HMM identifies facies by modeling hidden geological states based on sequential well log data. Transition probabilities capture facies relationship across depths. A mode-based moving window filter smoothed the initial results by reducing fluctuations, while preserving facies boundaries. Validation was performed by comparing the results with reference facies using accuracy, adjusted rand index (ARI), normalized mutual information (NMI), and visual assessment. Both HMM and KMC clustering revealed thin non-geological facies, primarily at the facies boundaries. In the middle of facies, HMM mitigated thin facies more effectively than KMC. These thin facies arose from fluctuations in the well logs. Visually, HMM provided more geologically realistic results, despite KMC sometimes showed slightly better performance metrics. To address thin artificial facies, a moving window filter was applied. This slightly improved clustering performance metrics but significantly enhanced geological consistency by removing most artificial facies at boundaries or within facies, particularly in the smoothed HMM. Smoothed HMM achieved slightly better or comparable accuracy and NMI to smoothed KMC. However, minor thin facies persisted in both methods, more notably in KMC, due to significant well log fluctuations. The filter effectively smoothed the facies by reducing variations in thin facies across small depth intervals, ultimately enhancing the facies continuity. This improved facies distribution coherency with geological expectation, particularly for the smoothed HMM. HMM with post-processing plays a crucial role in identifying geologically consistent facies while maintaining or slightly improving clustering performance metrics.

Unsupervised machine learning that incorporates sequential information is vital for facies clustering from well log data. The proposed workflow which integrates HMM with window filtering, enhances clustering

outcomes by producing more geologically consistent result and more interpretable facies. This approach aids geoscientists in reservoir characterization and modeling, enabling more informed decision-making.

Introduction

Accurate and reliable identification of subsurface geological facies is crucial for effective reservoir characterization. Facies analysis often relies on manual qualitative or supervised interpretations guided by expert geological insights, which can introduce subjectivity and inconsistency. Unsupervised machine learning methods, which do not require predefined labels, have become attractive alternative, allowing for objective and data-driven facies determination from well log data.

Among unsupervised approaches, KMC remains one of the most widely used algorithms due to its effectiveness in identifying distinct clusters. However, KMC does not inherently account for the sequential nature of well log data, and treats measurements independently (Lindberg and Grana 2015). In contrast, HMM are probabilistic models capable of capturing sequential dependencies within data, making them suitable for applications involving continuous depth-related data series, such as well logs (Rabiner 1989; Ephraim and Merhav 2002). HMM assumes that the observed log responses are manifestations of hidden facies states and explicitly model transition probabilities between these states. The inherent ability to incorporate stratigraphic continuity typically results in more geologically realistic facies predictions. Recent research has explored the application of HMM for well log data prediction, including lithology identification using HMM and random forest (Wang et al. 2020), logging data clustering for unit classification using HMM (Yabe et al. 2022), and petro-elastic log-facies classification using expectation maximization and HMM (Lindberg and Grana 2015). Nevertheless, even HMM may produce artificial facies characterized by thin intervals that typically are not geologically meaningful. This represents a significant challenge in facies prediction using machine learning approaches. These artificial thin facies intervals often appear at facies boundaries or within homogeneous geological layers. These facies commonly occur because of small scale fluctuations or noise in sensitive well log measurements. Such fluctuations compromise the geological interpretability of the clustering outcomes, posing significant challenges for geoscientists when translating clustering results into practical reservoir models. To mitigate these artificial thin facies intervals, post-processing techniques have been employed. Window filtering methods, specifically the mode-based moving window, have emerged as effective smoothing techniques that preserve critical geological boundaries while reducing fluctuations. By integrating window filtering techniques with clustering, the continuity and geological signatures of facies distributions can be significantly enhanced. This study examines the effectiveness of combining HMM with window filtering techniques for facies detection using synthetic well log data, including GR, DEN, and NEU logs. The primary objectives were to quantify the relative effectiveness of sequential versus non-sequential clustering methods in minimizing non-geological thin facies, and to assess the improvement in geological realism and consistency achieved through window filtering. Validation of clustering outcomes was performed through standard metrics such as accuracy, ARI, and NMI as well as qualitative visual assessments comparing the clustering results against interpreted geologist facies. Ultimately, the integrated approach presented herein aims to offer geoscientists a robust, reproducible, and unsupervised workflow for facies identification, facilitating improved reservoir characterization.

Methodology

Dataset

The main dataset utilized in this study comprised synthetic well log in stacking fluvial channel depositional environment. The dataset contained various well logs from 20 wells that simulate well logging conditions within gas reservoir condition. The dataset included the following 11 data : Well, DEPTH, Facies,

FaciesCode, gamma ray (GR), resistivity (RT), density (DEN), neutron (NEU), sonic (DT), porosity (PHIE), and permeability (PERM).

Facies

Geological facies represent rock bodies with a common set of properties relative to a common depositional environment (Reading 1996). On the other hand, electro-facies are classification of geological unit based on unique log responses derived from various well log measurements. This study focused on lithofacies identified from well log data, which reflect lithological and petrophysical properties. Typical well log data include GR, RES, DEN, and NEU data. GR measures formation radioactivity, with high values generally corresponding to shale-rich formations and lower values indicating sandstone. GR logs provide insights into lithology and correlations (Rider, 1996). DEN records bulk density of formations, helping to identify lithology, estimate porosity, and identify fluid. Low-density typically indicates porous reservoir rocks, while high-density suggests compact formations such as shale. NEU measures hydrogen content, which is related to fluid-filled porosity. Lower neutron corresponds to porous formations containing hydrocarbons or water. RES measures formation resistivity, which is resistance to electrical current. High resistivity usually shows hydrocarbons, while low resistivity suggests water-filled formations or shales. Distinctive log response patterns may correspond to specific depositional environments (Fig. 1). Channel deposits are indicated by blocky to fining-upward GR log. Deltaic systems typically exhibit coarsening-upward or funnel-shaped GR patterns that reflect delta progradation.

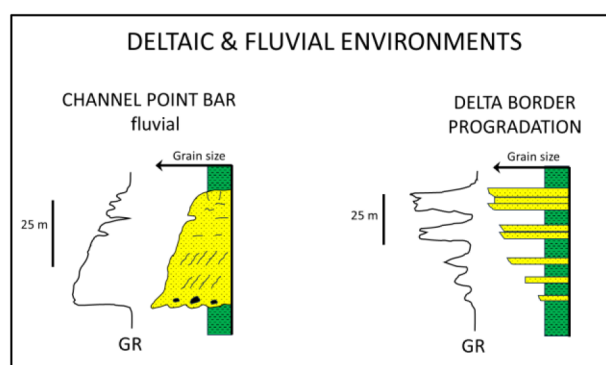


Figure 1—GR response and depositional setting interpretation (modified from Rider, 1996)

Workflow

The proposed workflow integrates HMM with a smoothing step based on window filtering (Fig. 2). For comparison, a non-sequential clustering KMC was also applied, with both methods evaluated before and after filtering.

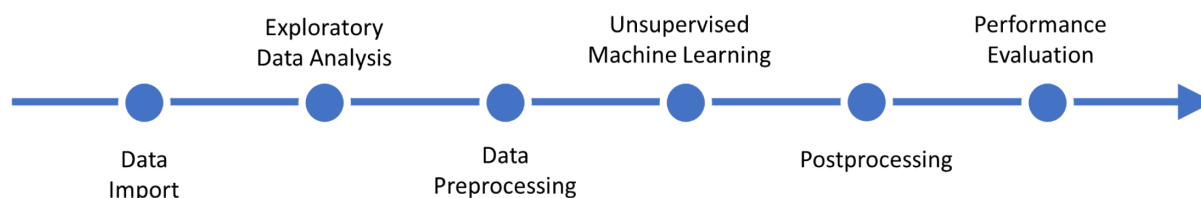


Figure 2—General workflow of the methodology

Data Import

The well log data were loaded and prepared for analysis. The dataset available in this study consists of a combined synthetic well log from 20 wells as a structured dataset, including well, depth, GR, DEN, NEU, RT, DT, PHIE, PERM, and facies code.

Exploratory Data Analysis (EDA)

EDA commenced with inspection of dataset to understand its structure, including variable types, dimensionality, and initial entries. Univariate analysis was conducted to understand the distribution of individual features, whereas bivariate and multivariate analyses, including scatter plots and correlation matrices, were used to identify inter-variable relationships. Based on the findings, preprocessing steps were performed.

Data Preprocessing

The final dataset used in this study comprised GR, DEN, and NEU logs. These logs were normalized using statistical standardization to ensure uniform scaling across features, which is a critical step in enhancing the performance of machine learning algorithms.

Unsupervised Machine Learning

K-Means Clustering (KMC). KMC is an unsupervised algorithm that partitions data points into distinct clusters by minimizing the variance within each cluster (Bhattacharya 2021) (Fig. 3). KMC was employed as the baseline algorithm. For this study, four clusters were defined to match the ground truth facies number. The procedure involved in KMC consist of repeated processes (Belyadi and Haghighat 2021) such as initialization of cluster centroids randomly or intentionally, assignment of data points to the nearest centroid, updating the centroid positions based on the assigned data points, and iterative recalculation of centroids until convergence.

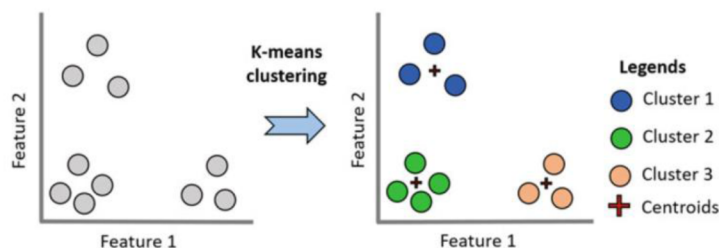


Figure 3—Simplified concept of KMC algorithm (Bhattacharya 2021)

Hidden Markov Model (HMM). HMM is a stochastic process used to model systems where the underlying states are not directly observable. HMM is used to infer the most probable sequence of hidden states, given a sequence of observations (Fig. 4a). HMM can address the sequential nature of well log data. HMM treats observations (e.g., well logs) generated by a sequence of hidden states as emissions (e.g., facies), which cannot be directly observed but can be inferred through observed data (Fig. 4b). The main components of an HMM include (Alpaydin 2020; Rabiner 1989): 1) Hidden states $S = \{s_1, s_2, \dots, s_T\}$ represent discrete geological facies; 2) Observable variables $O = \{o_1, o_2, \dots, o_T\}$ cover actual measurements from well logs (eg. GR, DEN, NEU); 3) A transition probability matrix $A = [a_{ij}]$ where $a_{ij} = P(s_j | s_i)$. Transition probabilities signify the transition from one facies to another, capturing geological continuity; 4) An emission probability matrix $B = [b_j(O_t)]$ where $b_j(O_t) = P(O_t | s_j)$. Emission probabilities handle the probability distribution of the observed data for each given facies; 5) An initial state distribution $\pi = [\pi_i]$, where $\pi_i = P(s_i \text{ at } t=0)$. The initial state probabilities imply likelihood of starting in any given facies. Where π = initial hidden state probability, S = hidden state, O = Observation, A = hidden to hidden transition, B = hidden to observation emission, P = probability

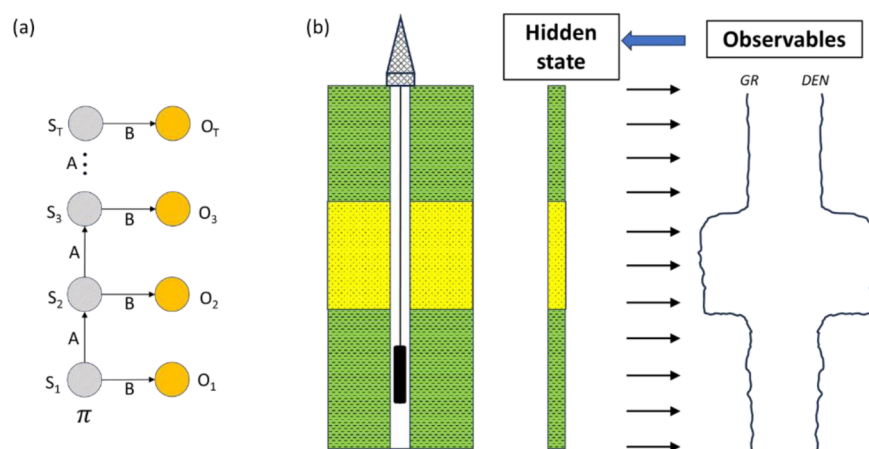


Figure 4—(a) Simplified concept of HMM algorithm (modified from Jordan et al. 1999); (b) Schematic concept of HMM for well log data (modified from Yabe et al. 2022)

The transition matrix was designed with a bias toward self-transitions, simulating the facies trend over depth. Training was performed iteratively to optimize the model parameters using the expectation-maximization algorithm, and the most probable facies sequence was inferred using the Viterbi algorithm (modified from Alpaydin 2020; Rabiner 1989). The HMM procedure involves several steps (Alpaydin 2020; Rabiner 1989). Evaluation step involves computing the probability of an observation sequence. The likelihood is calculated using the forward algorithm, which recursively accumulates the probabilities. Decoding step focuses on determining the most probable sequence of hidden states. The Viterbi algorithm is utilized, to find the optimal path through the state space that corresponds to the observed data. Learning stage includes estimating the model parameters. With only observation sequences, the model is trained by iteratively adjusting these parameters to maximize the likelihood. The optimization is performed using expectation-maximization algorithm.

Post-Processing

Mode-Based Window Filtering. This filtering involves sliding a fixed-size window across the sequence of data and replacing each point with a specific statistical value of the data within that window (modified from Virtanen et al. 2020). The selection of the window size is critical. A smaller window retains more detail but may not sufficiently reduce noise, whereas a larger window might over smooth and eliminate meaningful transitions. In this study, 10 window size was determined based on log sampling interval and empirical testing to balance facies realism and noise reduction.

Label Alignment and Mapping. A known challenge in unsupervised learning is the arbitrary assignment of the output cluster labels, which may not correspond to the reference labels used for validation. To fairly evaluate model performance, a label alignment procedure was implemented. This technique ensures optimal one-to-one mapping between the predicted and true labels (Kuhn 1955).

Performance Evaluation

Comparisons between the prediction result and the original facies were performed for performance evaluation using quantitative and qualitative approaches. Each cluster output was quantitatively evaluated using three metrics. Accuracy measures the number of data points classified correctly that provides a general indicator of the overall clustering accuracy (Belyadi and Haghighat 2021). ARI evaluates the similarity between predicted and real clustering where -1 indicates no agreement and 1 indicates perfect agreement (Hubert and Arabie 1985). NMI measures mutual dependency between predicted and true cluster facies, which is normalized for scale to avoid bias in cluster sizes where 0 means no mutual information and 1

means perfect correlation (Vinh et al. 2010). In addition, visual assessment complemented these metrics. The results were visualized and compared using well log display or scatter plots. Visual inspection assesses facies continuity and boundary coherency, a characteristic typically not measured by numerical metrics but fundamental for reservoir characterization.

Results and Discussion

Exploratory Data Analysis (EDA)

Descriptive statistics were generated for all numerical features to analyze the statistical characteristics (Table 1). Univariate histograms revealed skewness and multimodality distributions for each data, indicating heterogeneity and possible nonlinear behavior (Fig. 5). Pair plots were constructed to visualize the facies distribution and as indication of relationship between features (Fig. 5). These pair plots revealed overlaps and separability between facies classes and provided insights into which features were the most discriminative. Notably, GR, as the primary lithology log, exhibited a strong correlation with DEN and NEU, which is consistent with common lithology responses, followed by DT and RT in inverse correlation. PHIE and PERM, as interpreted logs, had a moderate inverse correlation with GR. These findings serve as a foundation for subsequent facies classification and reservoir property prediction tasks.

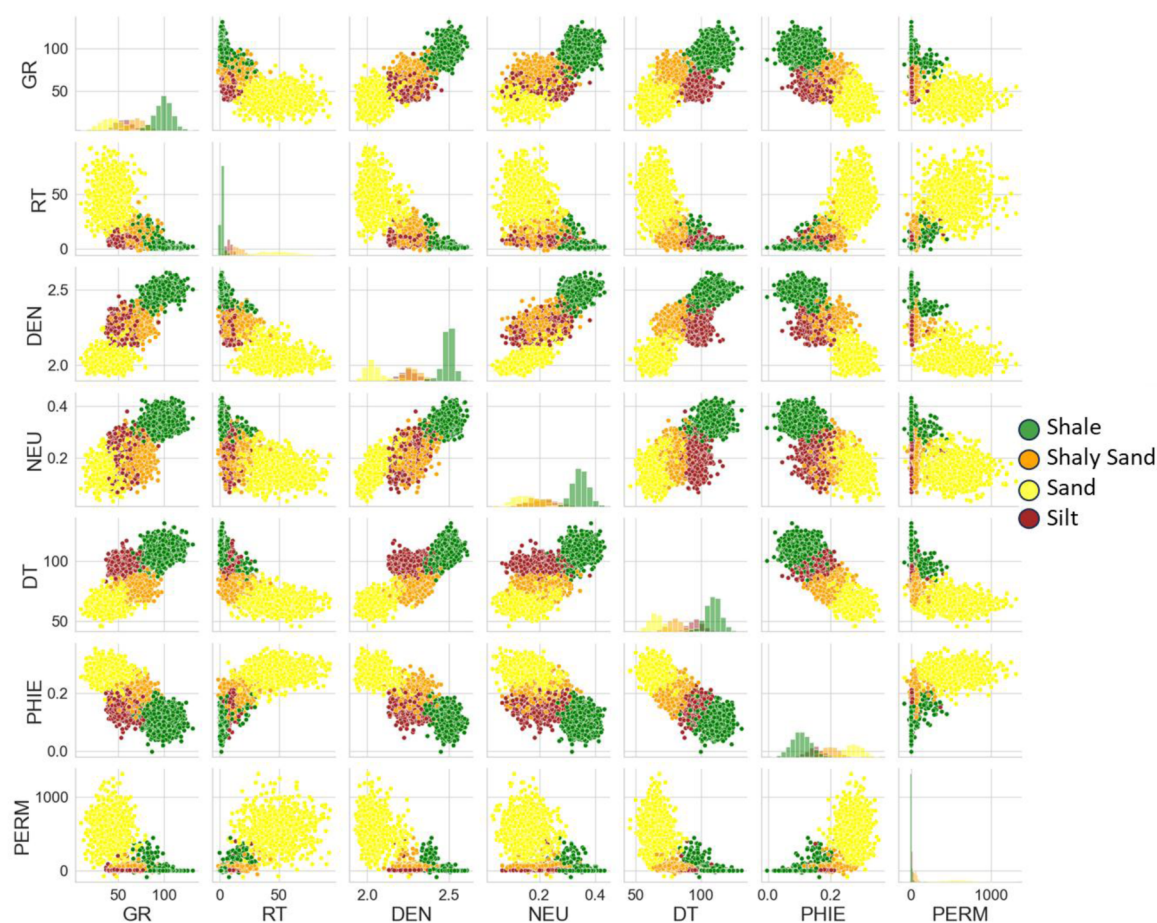


Figure 5—Pair plot between input features overlain with facies

Table 1—Descriptive statistic of main dataset

	dtype	count	mean	std	min	25%	50%	75%	max
DEPTH	float64	3600	1044.8	26.0	1000	1022.375	1044.75	1067.125	1089.5
FaciesCode	int64	3600	2.8	1.2	1	2	3	4	4
GR	float64	3600	74.6	25.6	9.7	53.1	75.2	97.8	131.0
RT	float64	3600	15.9	20.3	-1.3	1.7	6.8	20.8	93.2
DEN	float64	3600	2.3	0.2	1.9	2.2	2.3	2.5	2.6
NEU	float64	3600	0.3	0.1	0.0	0.2	0.3	0.3	0.4
DT	float64	3600	92.2	18.8	45.5	75.6	97.0	108.4	131.2
PHIE	float64	3600	0.2	0.1	0.0	0.1	0.2	0.2	0.4
PERM	float64	3600	145.2	254.6	-86.9	0.6	11.0	97.9	1313.2

Clustering

GR, DEN, and NEU logs were selected as the final input features for clustering due to their superior ability to distinguish between common lithologies such as sand, shale, siltstone, and shaly sand. RT is very sensitive to fluid, whereas DT is a kind of special well log that will not be acquired for all standard wells. PHIE and PERM are interpreted logs that are not available directly when drilling a well. 4 clusters were defined to match the number of ground truth facies interpreted by geologist that consist of sand, shaly sand, siltstone, and shale. The clustering results obtained using KMC and HMM, both before and after smoothing with window filtering, were evaluated in terms of quantitative metrics and facies consistency and continuity (Fig. 6). Each clustering result was compared with the ground truth facies using three quantitative metrics which include accuracy, ARI, and NMI (Table 2).

Table 2—Clustering performance metrics table

Algorithm	Accuracy	ARI	NMI
KMC	0.813	0.789	0.692
Smoothed KMC	0.844	0.804	0.702
HMM	0.813	0.765	0.707
Smoothed HMM	0.847	0.797	0.716

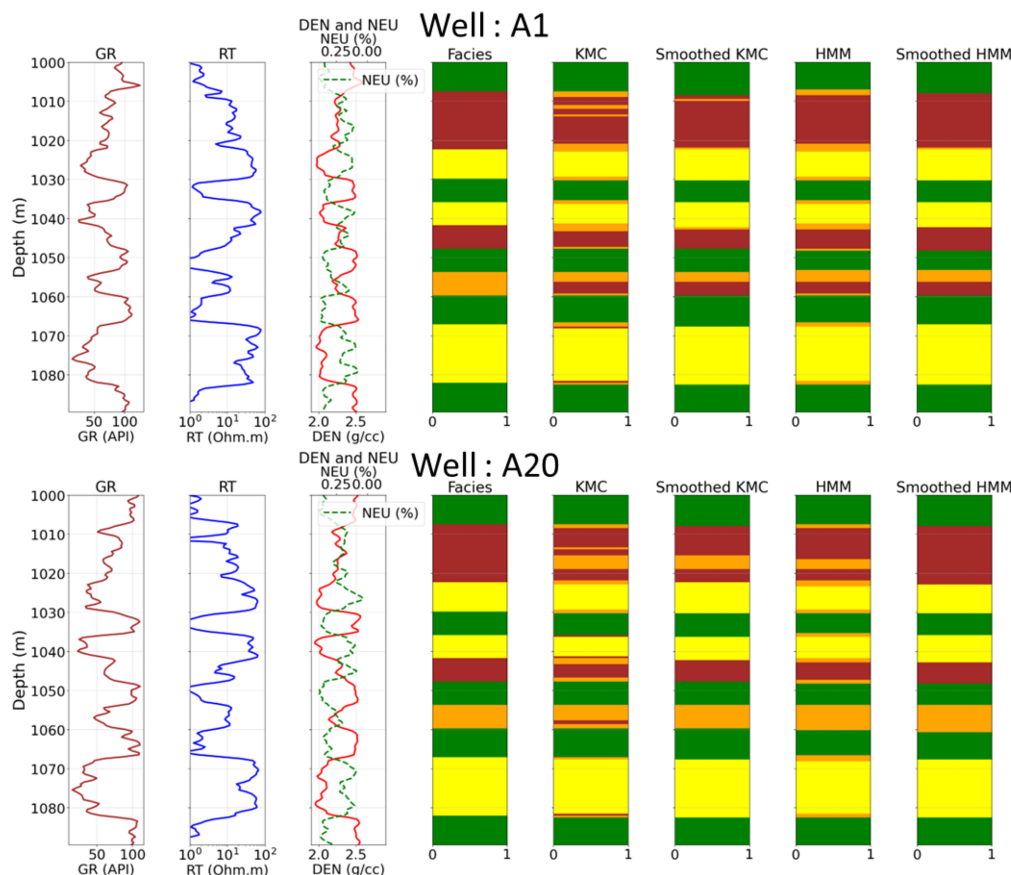


Figure 6—Example of well log visualization between input data and result of clustering

The KMC method achieved slightly higher accuracy and ARI than the HMM before smoothing, indicating that it performed better in terms of statistical matching with the true facies labels. However, the accuracy of the HMM was competitive, and slightly higher in NMI, and its visual results were more geologically reasonable. After smoothing with the mode-based moving-window filter, the performance of both methods improved slightly in all metrics. The smoothed HMM, in particular, demonstrated improved facies continuity while maintaining or improving the quantitative performance, as revealed by slightly higher accuracy and NMI compared to smoothed KMC. Smoothed HMM showed slightly better separation between facies classifications compared to smoothed KMC, as shown mainly in the DEN and NEU pair plot and histogram of NEU overlain with the smoothed KMC and HMM facies (Fig. 7).

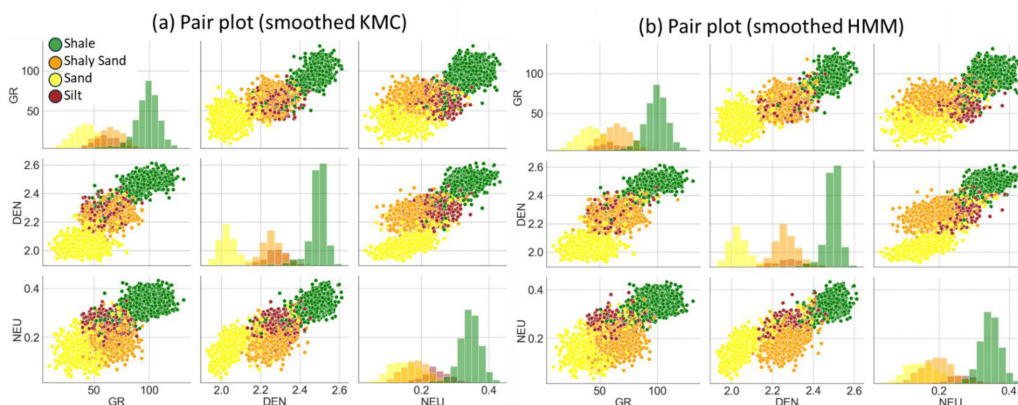


Figure 7—Pair plot between input features overlain with facies clustering result, smoothed KMC (a) and smoothed HMM (b)

Thin Facies Artifacts

Thin facies intervals that span only a few depth samples pose significant challenges in both the KMC and HMM clustering outputs. In the case of KMC, these artifacts are primarily driven by local log variability because each depth point is treated independently (Lindberg and Grana 2015). The general facies features were captured using the KMC approach but revealed non-geological thin facies, either in the middle or at the boundaries of the facies that related to fluctuations in well log, particularly GR and NEU (Fig. 6). The HMM addresses this issue more effectively by favoring transitions between similar states. HMM, which accounts for sequential relationships between data points, produced improved facies predictions by mitigating vertical facies continuity. Facies continuity using HMM has also been observed in previous studies (Lindberg and Grana 2015). However, due to the probabilistic nature of emission distributions and the presence of noise, some thin facies intervals still emerged in the initial HMM predictions, particularly near the facies boundaries (Fig. 6). A moving-window filter with size of 10 was applied to address thin artificial facies. The filter worked by collapsing short noisy transitions into the dominant facies of the surrounding interval. This not only improved the visual quality of the facies log, but was also more consistent with the expected geological facies patterns, particularly in fluvial channel environments. Applying a window-based mode filter effectively reduced the presence of noise-induced thin, non-geological facies intervals, while preserving significant geological boundaries, resulting in smoother and more continuous facies distributions. The smoothed KMC method was able to visually enhance facies prediction. The thin middle facies can be smoothed intensely, but thin facies still occur at the facies boundary. Compared to smoothed KMC, the filtered and smoothed HMM significantly improved facies clustering by reducing and removing most of the thin facies both in the middle and at the border facies. The smoothed HMM performed particularly well in the interior of major facies zones, where it was able to maintain the dominant lithology and minimize noise-induced transitions. At facies boundaries, the combination of HMM and filtering provided a more geological representation than KMC, which often failed to distinguish transitional zones. Even though the smoothed HMM significantly reduces the thin facies across small depth intervals, minor thin facies intervals can still be found, particularly in the facies boundaries, mainly due to significant well log variation (Fig.6). HMM with window filtering improves visual and geological interpretability makes it a more appropriate tool for facies detection in sequential well log data. By improving the geological consistency of facies predictions, the proposed method enhances the interpretability and utility of unsupervised machine learning outputs.

Conclusion

This study demonstrates that integrating HMM with window filtering is an effective approach for unsupervised facies detection using well log data. GR, NEU, and DEN logs representing channel depositional facies with geologist facies labels were used. The HMM with sequential characteristics generated more consistent facies sequences than the non-sequential KMC. The KMC frequently introduced non-geological thin facies, although it occasionally achieved marginally better numerical metrics. The quality of the HMM and KMC improved significantly with window filtering by eliminating the thin artificial facies intervals. This process enhanced the facies continuity without sacrificing performance metrics. An optimal balance between numerical accuracy and facies interpretability can be achieved by smoothed HMM, where it matches or outperforms smoothed KMC in numerical metrics while providing superior facies consistency. The predicted facies from this approach serves as valuable inputs and assist geoscientists in reservoir characterization, reservoir modeling, and well operation.

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